

Systems of Linear Equations

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$$3a + 2b + c = 0$$

$$7a + 6b + 5c = 2$$

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Inconsistent → System has no solution

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System has multiple solutions

$$\begin{bmatrix} 7 \\ 2 \end{bmatrix}$$

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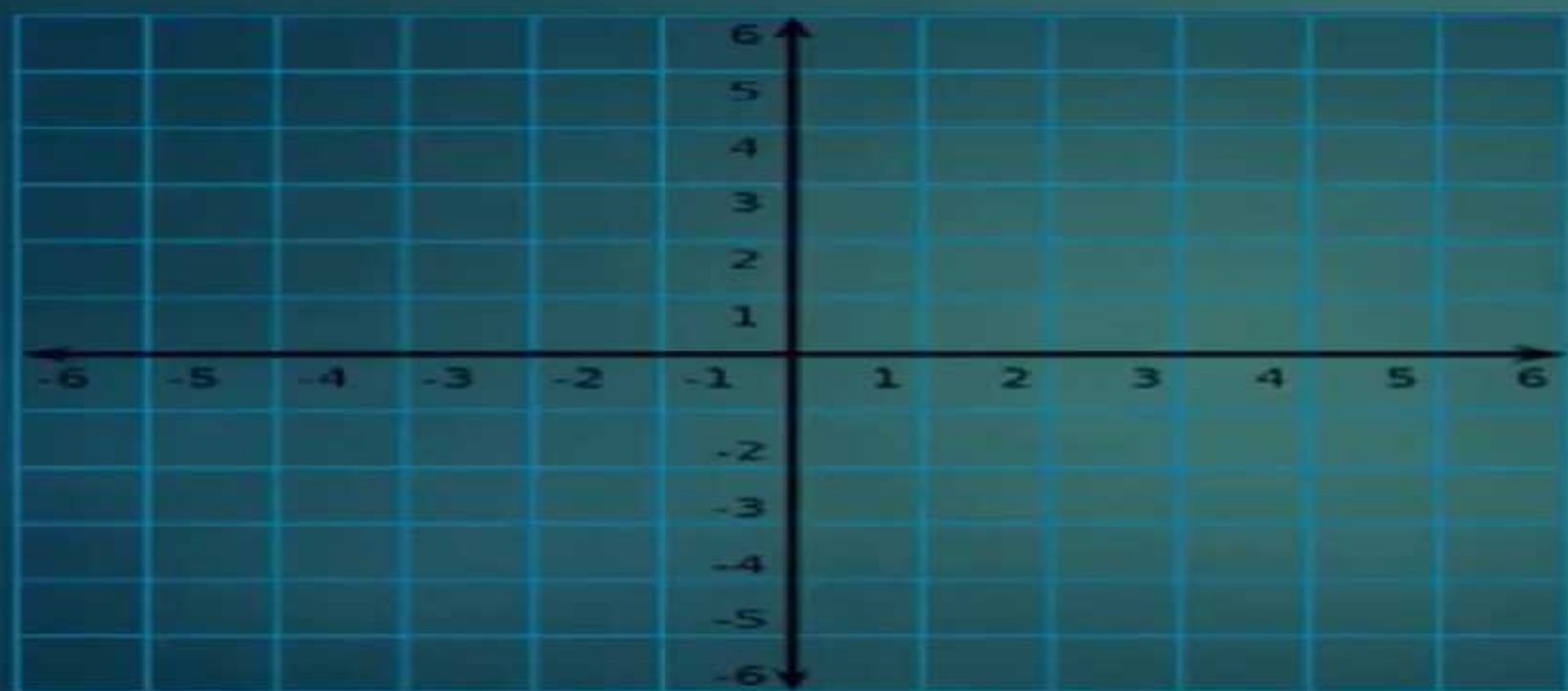


$$\begin{bmatrix} 7 \\ 2 \end{bmatrix}$$

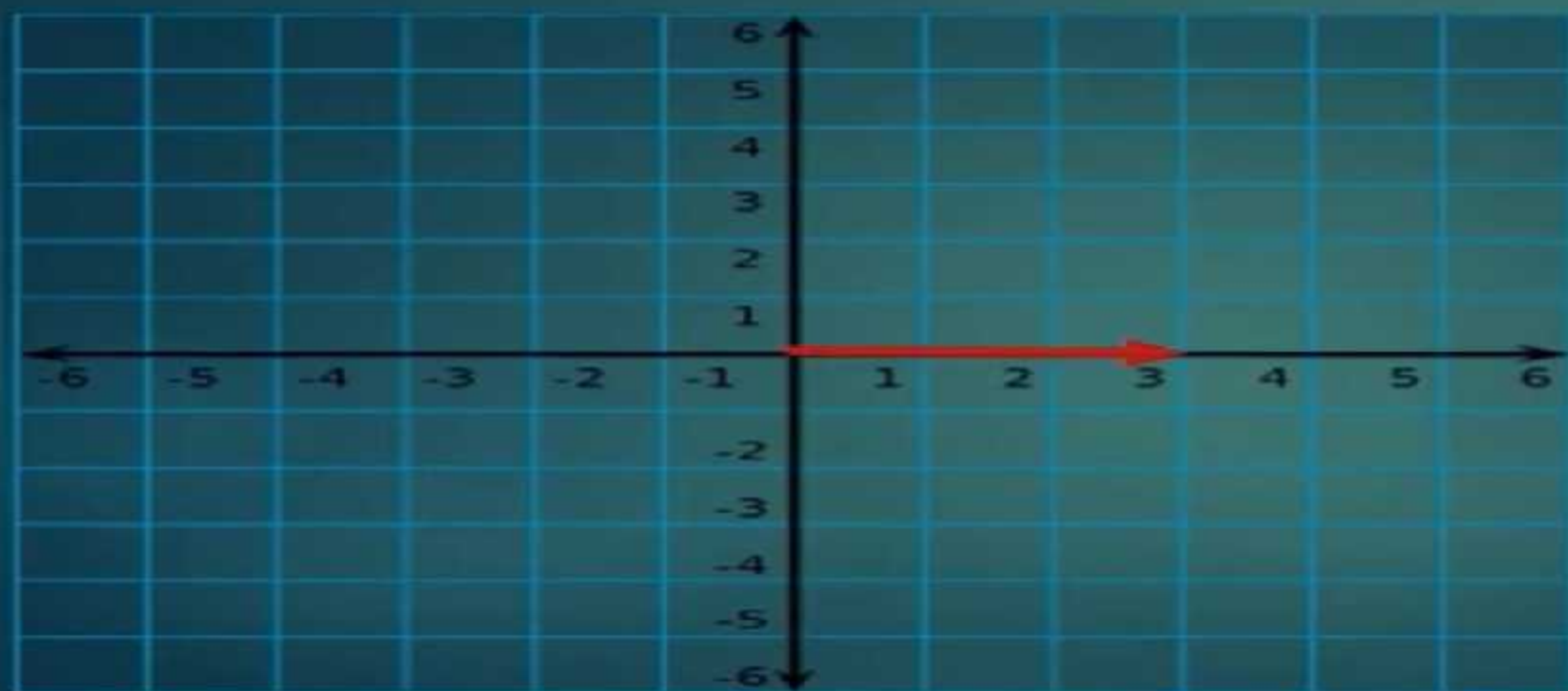


$$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

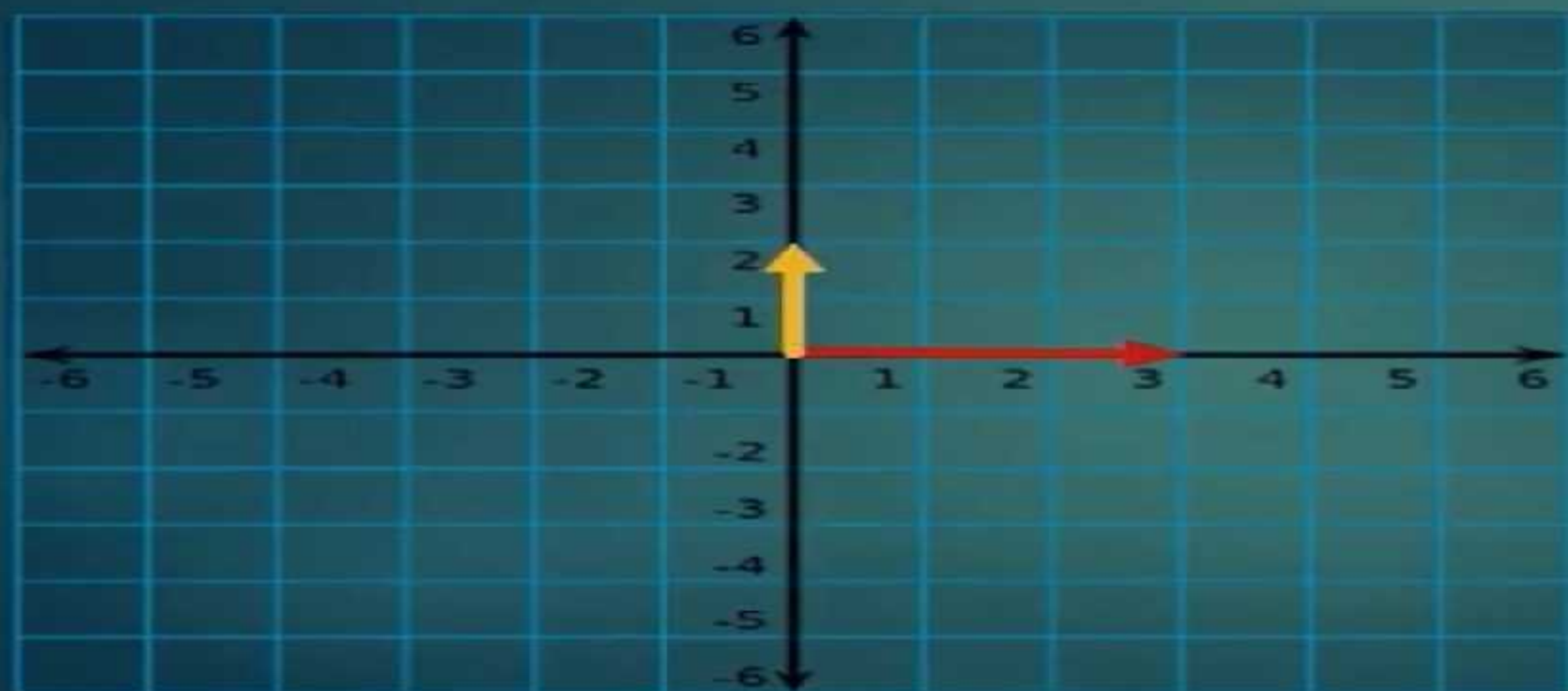
$$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$



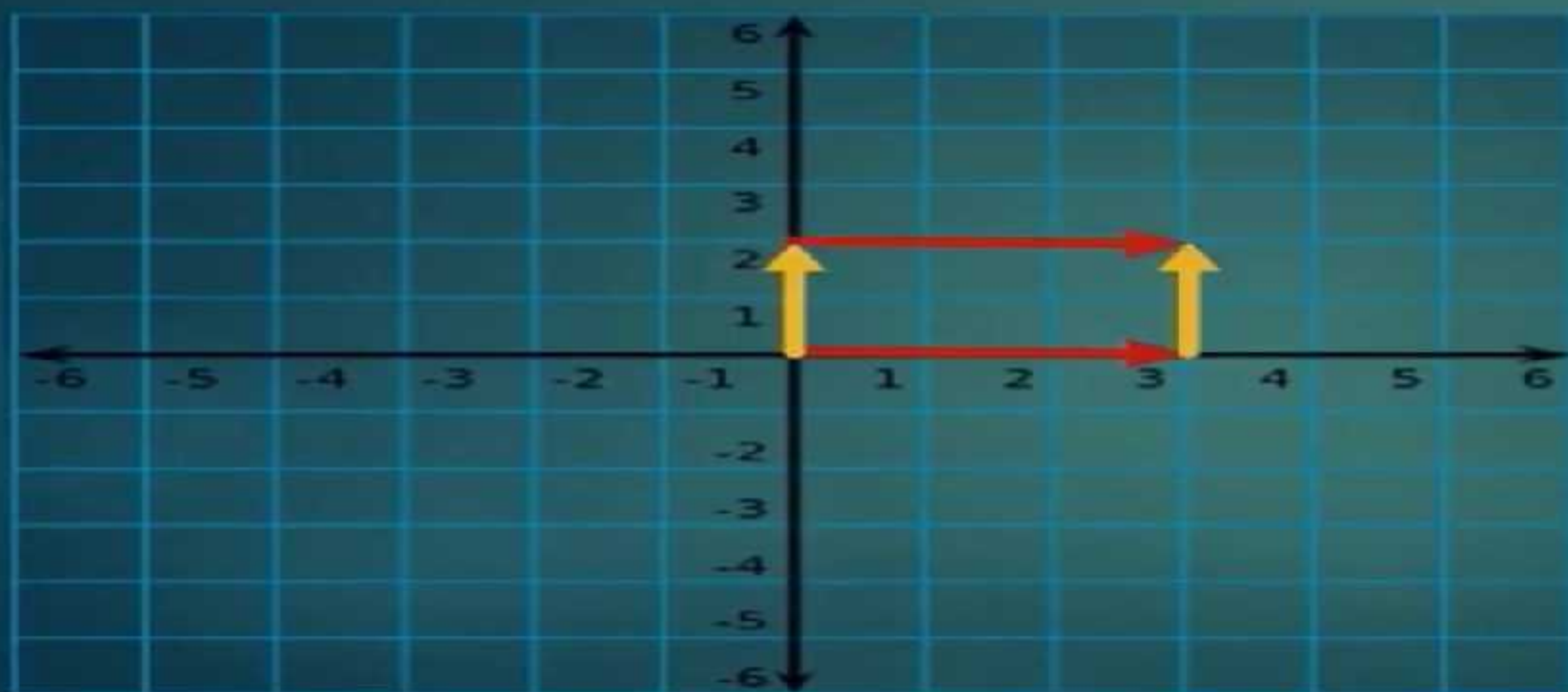
$$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$



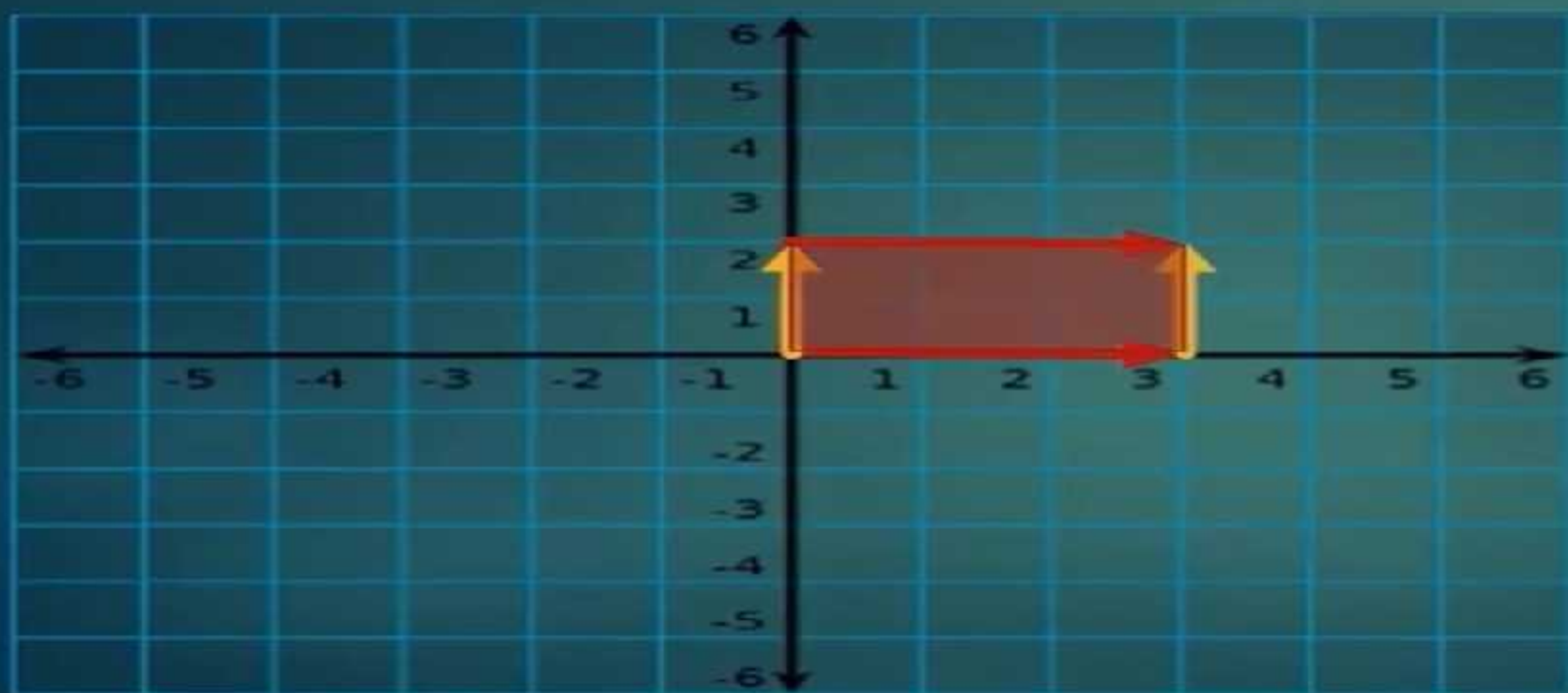
$$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$



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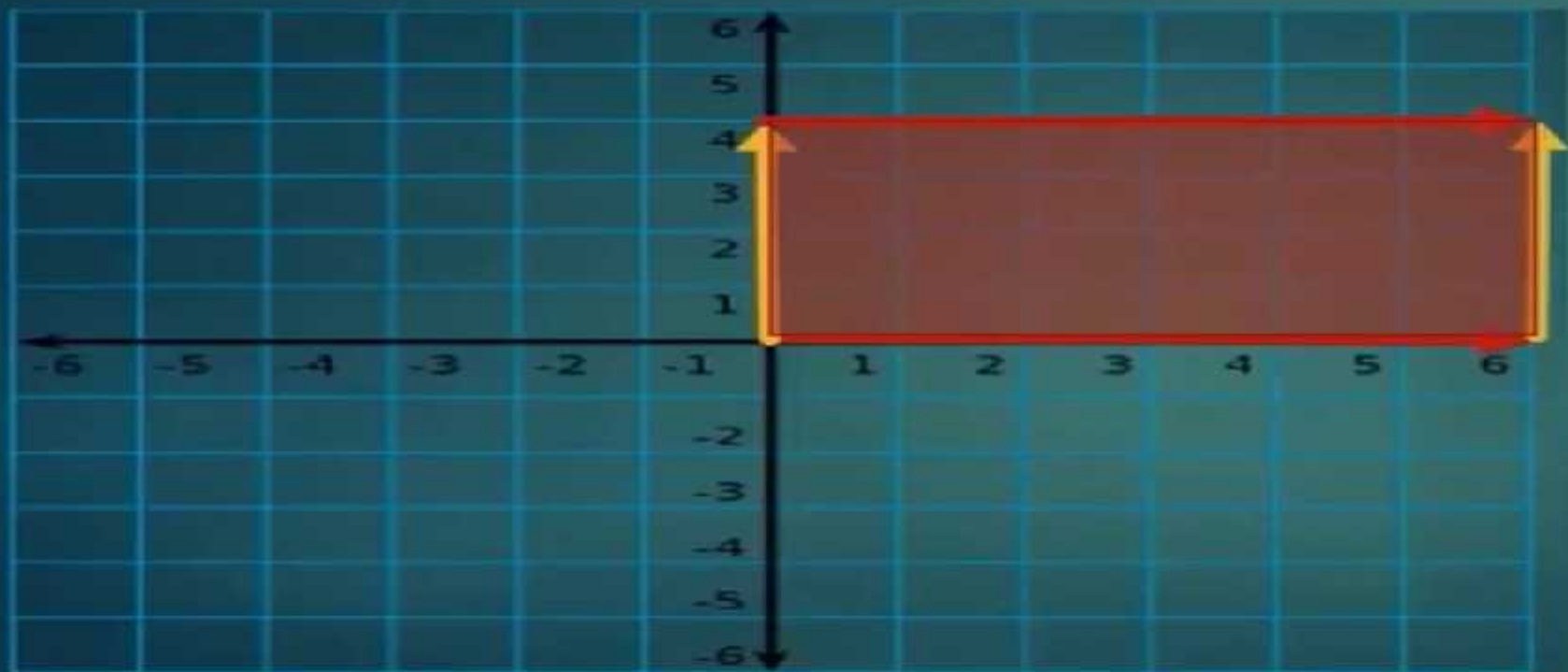


$$2 \times \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

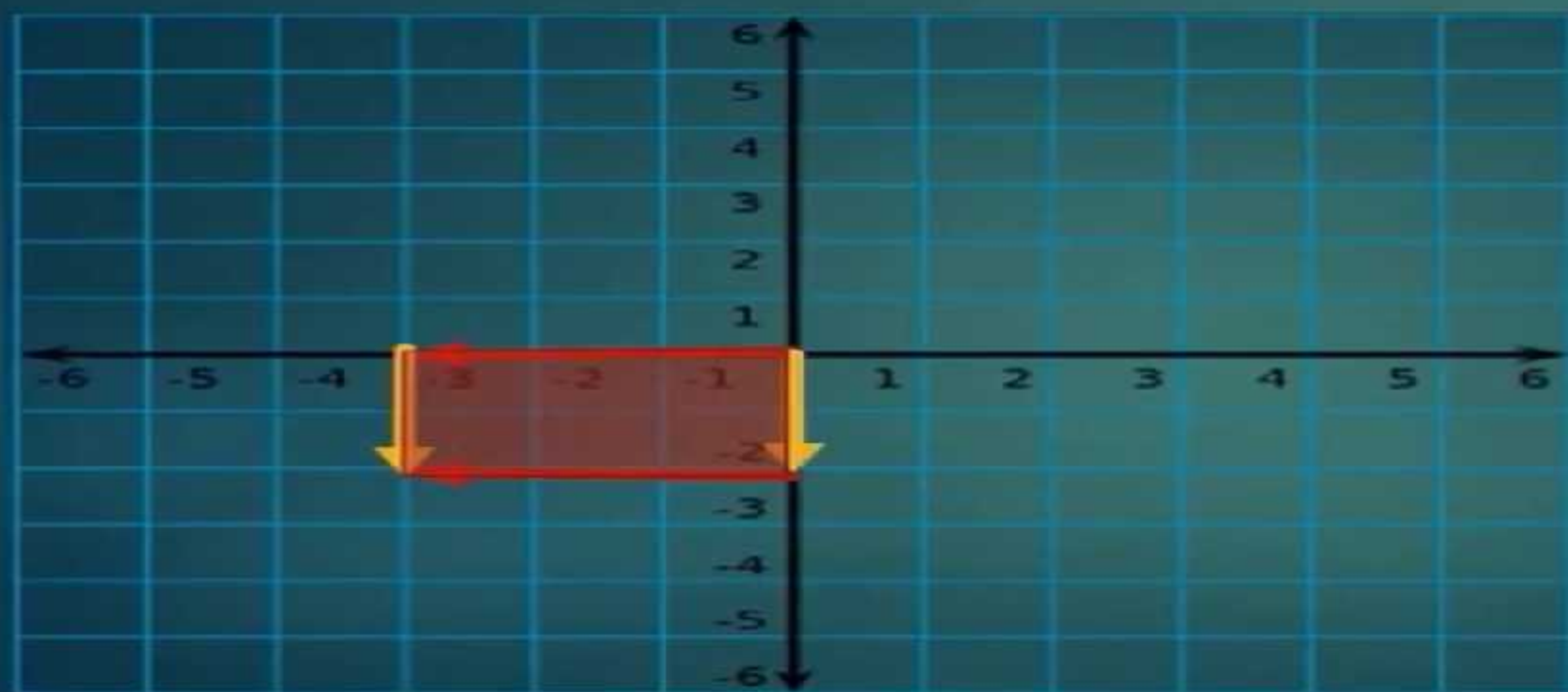
$$2 \times \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 0 \\ 0 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 0 \\ 0 & 4 \end{bmatrix}$$



$$\begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

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Linear Transformation

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1 - Scale

Linear Transformation

1- Scale

2- Rotation

Linear Transformation

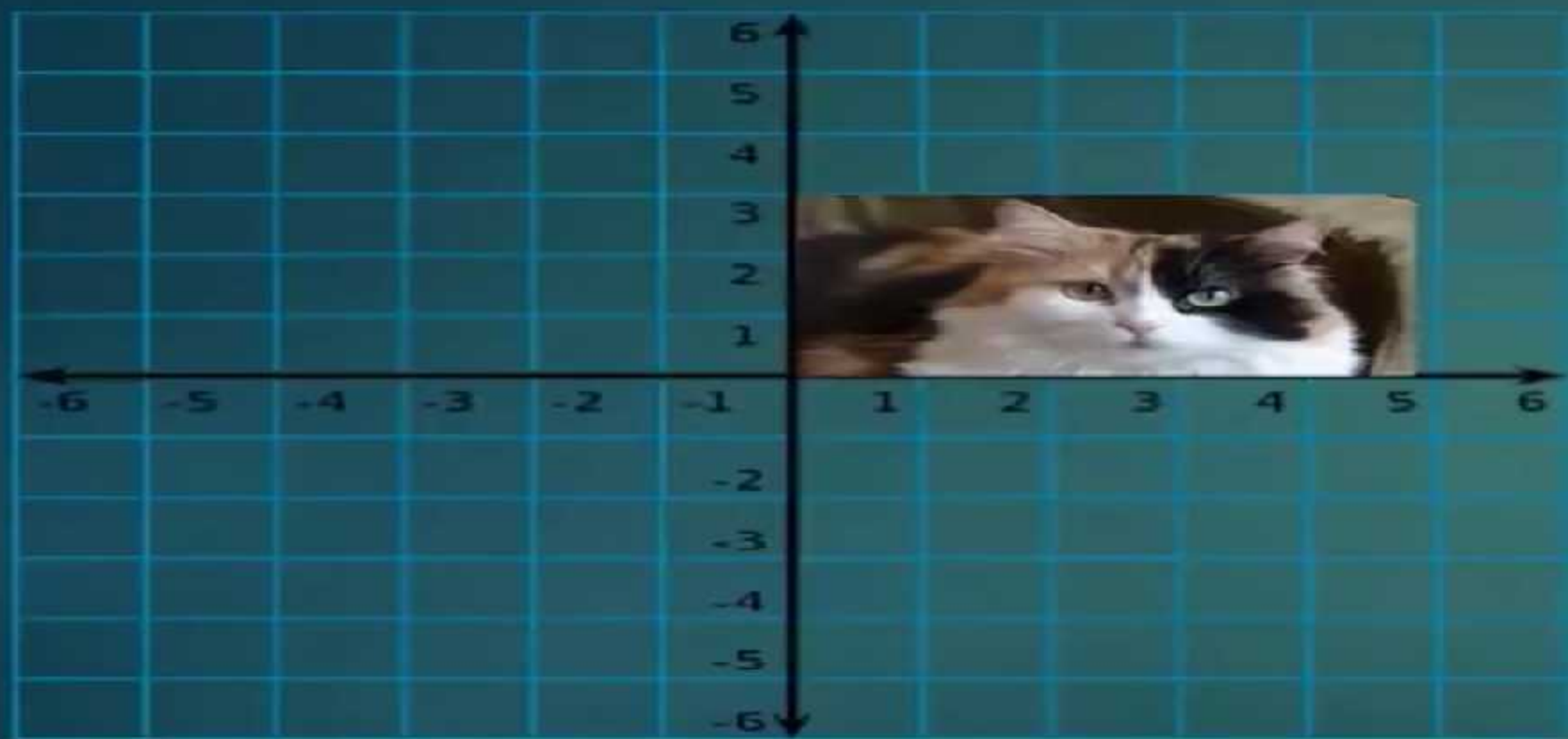
1- Scale

2- Rotation

3- Flipping

Linear Transformation

- 1- Scale
- 2- Rotation
- 3- Flipping
- 4- Skew

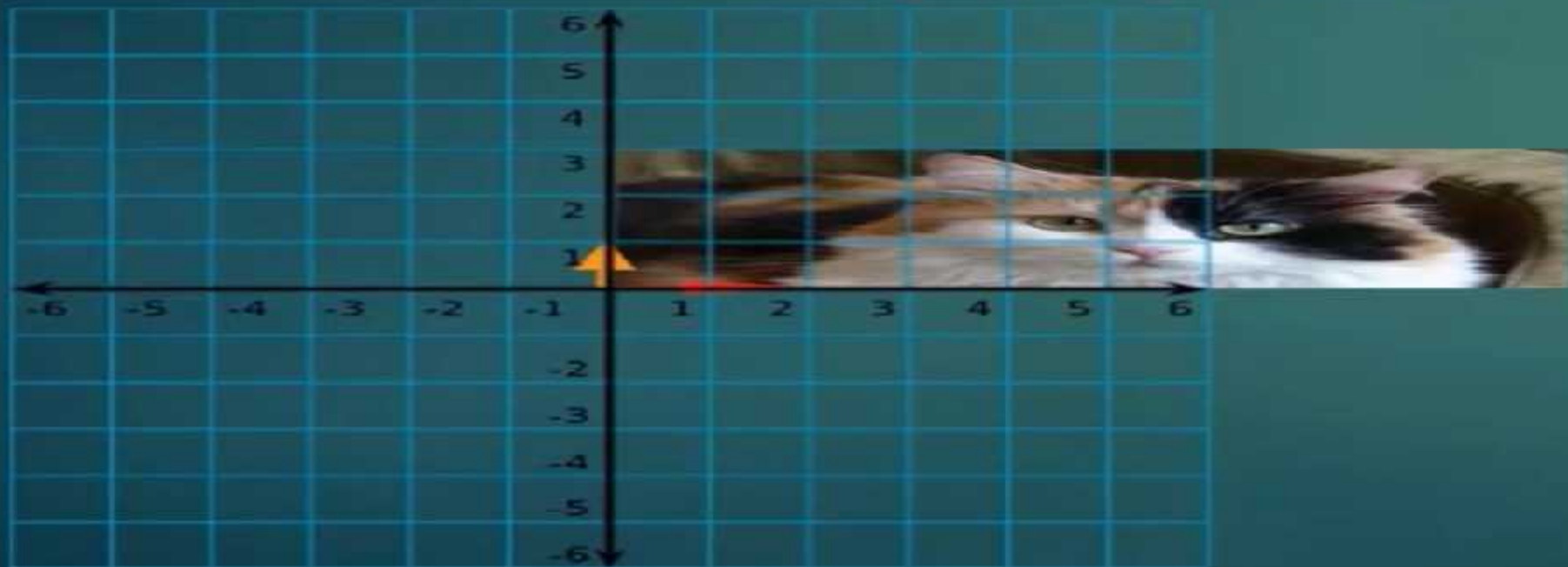




$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}$$



$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}$$



$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{x}$$



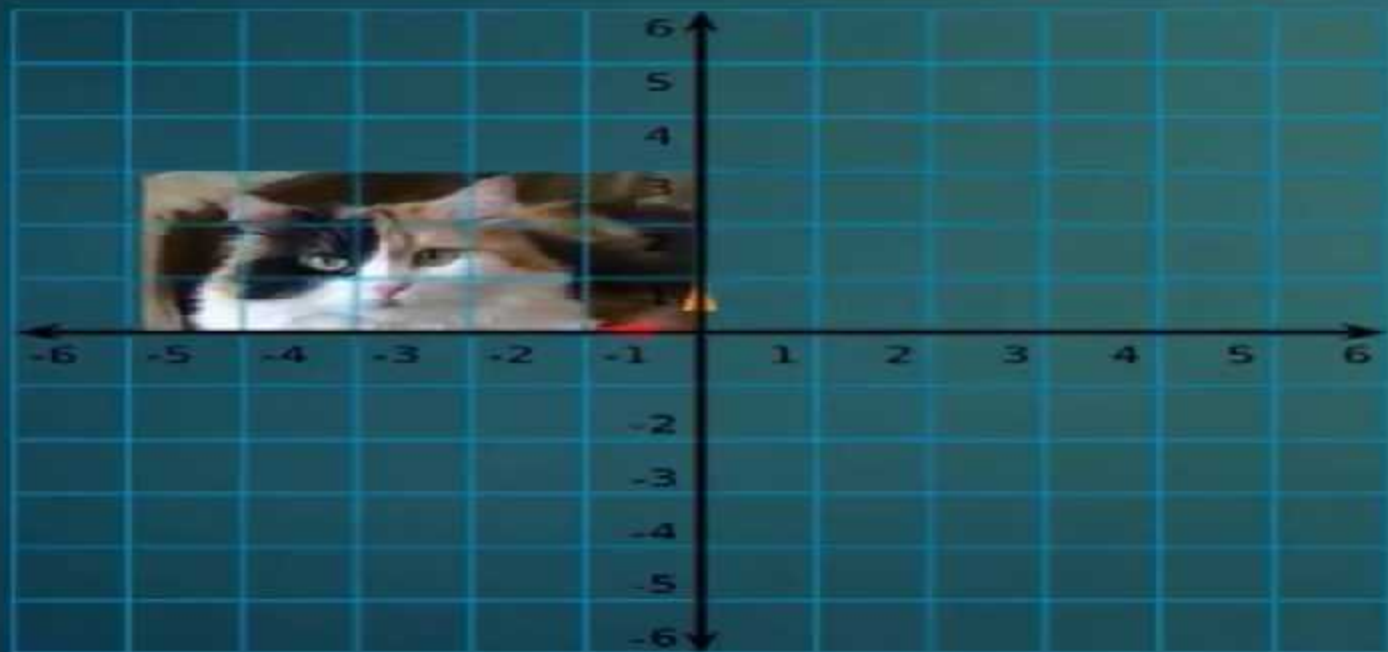
$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{x}$$



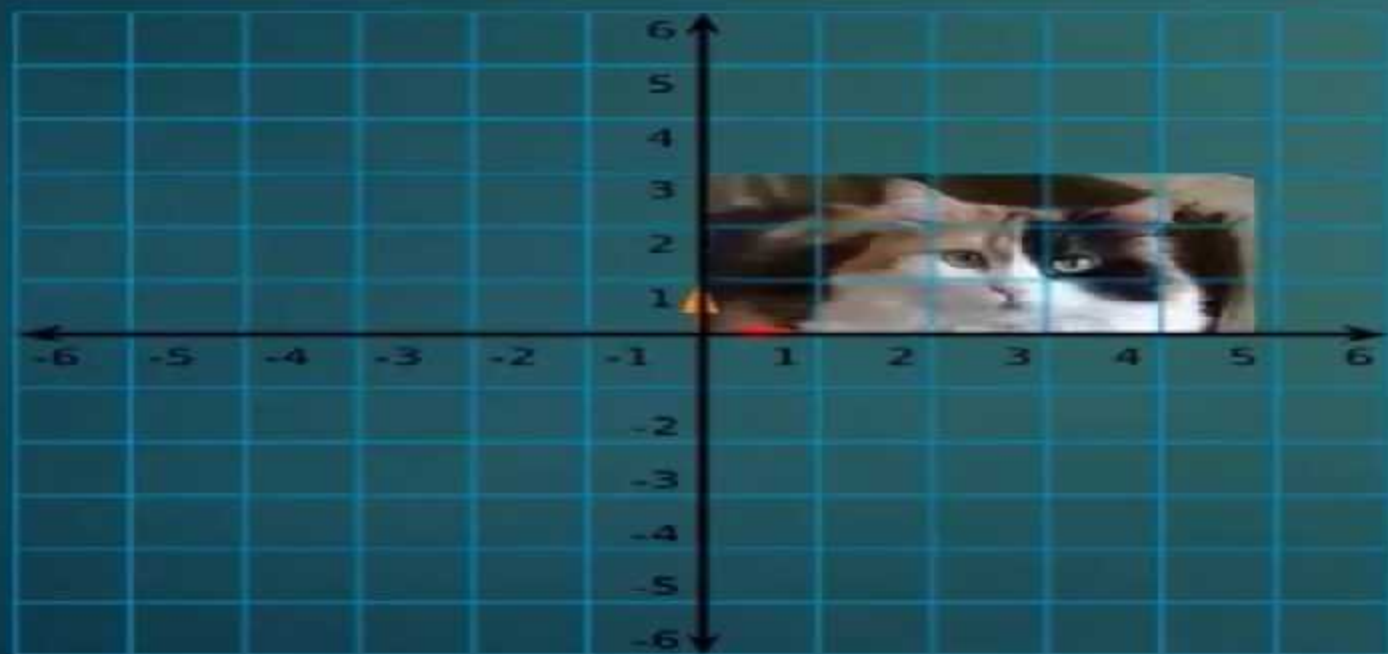
$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}$$



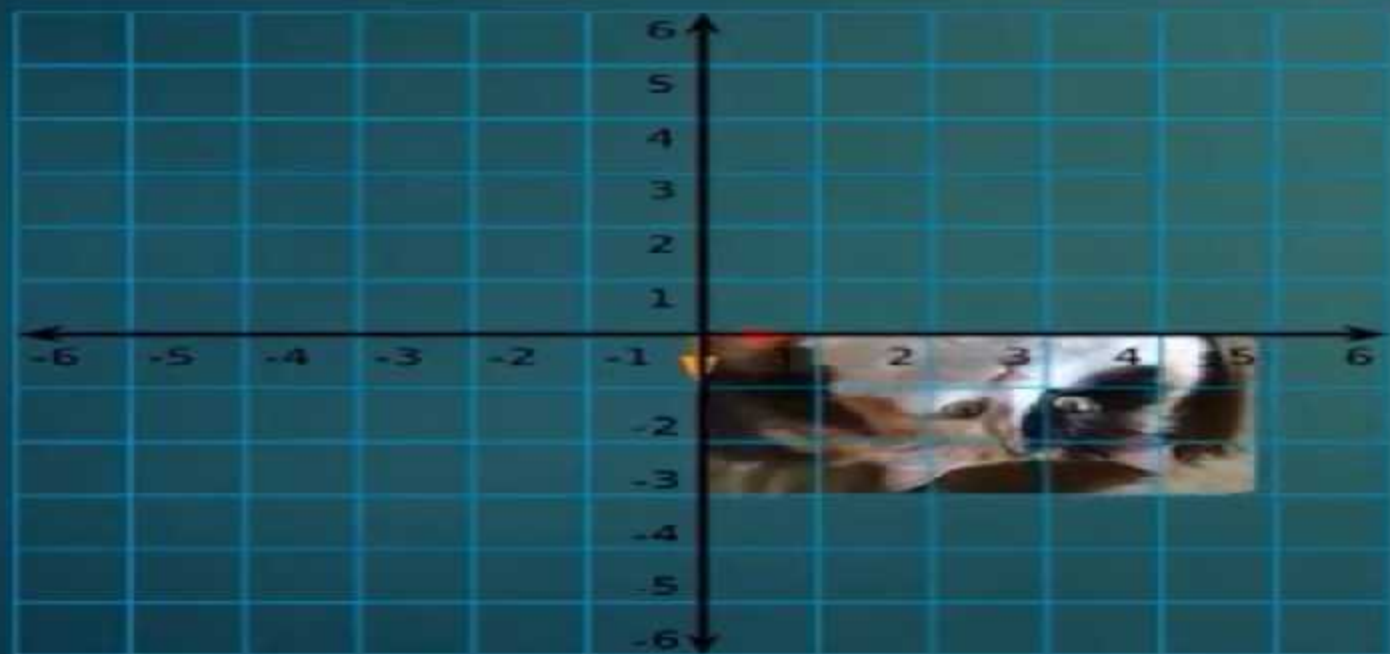
$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}$$



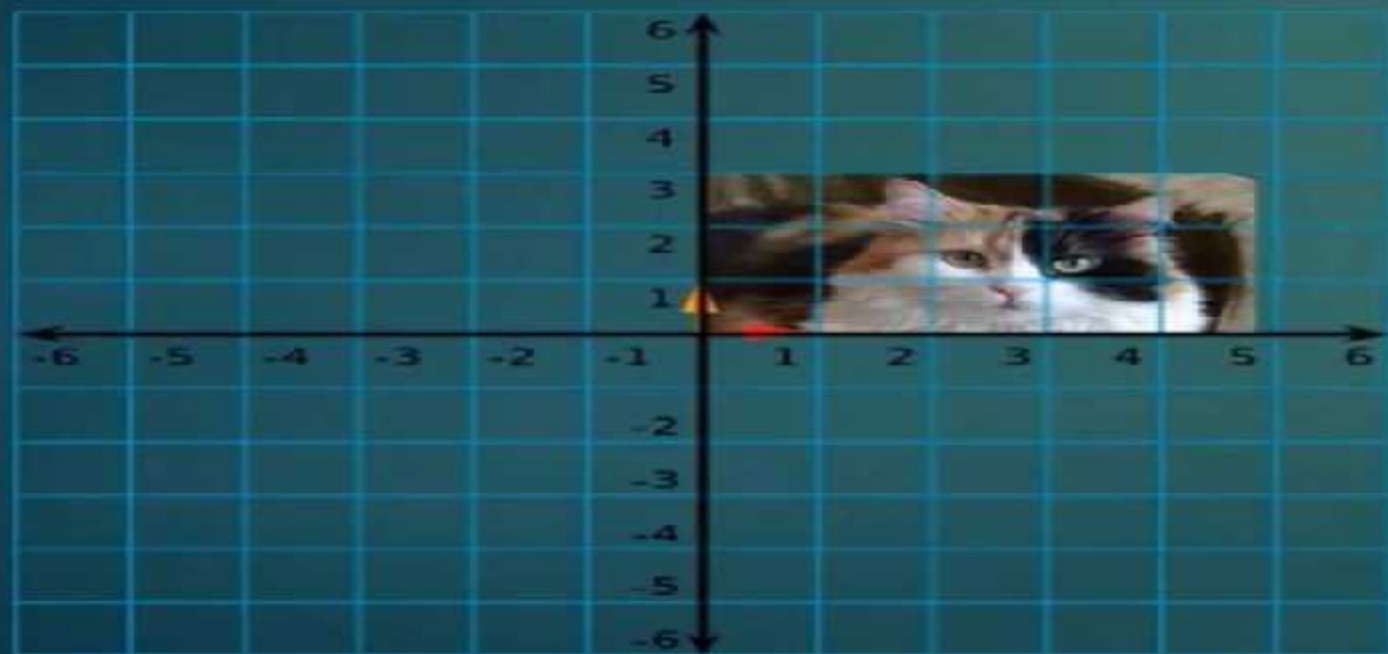
$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x}$$



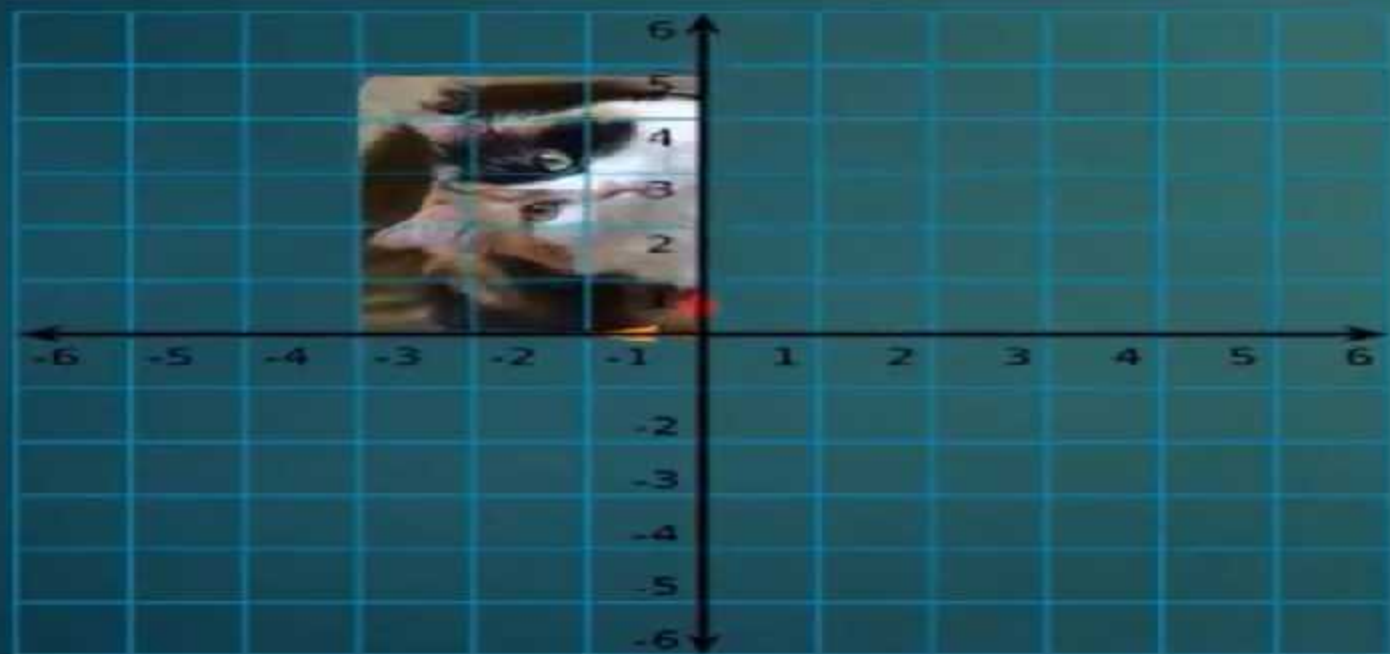
$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x}$$



$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} x$$



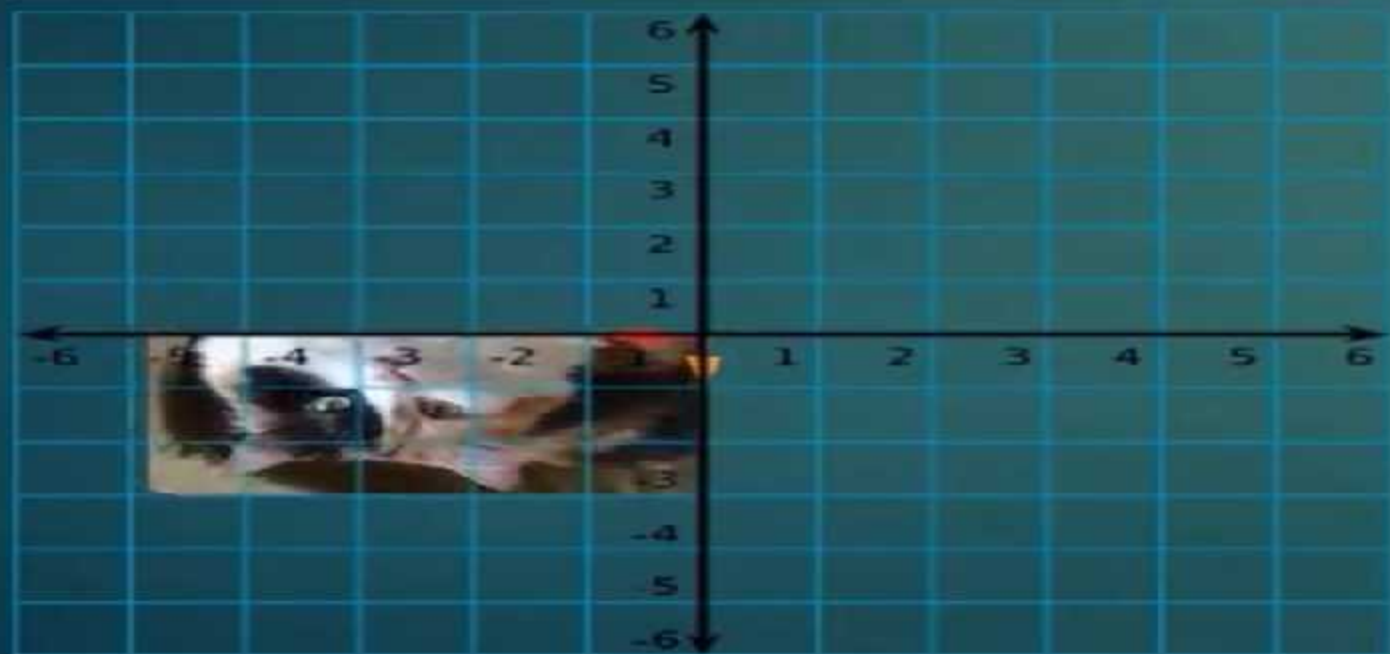
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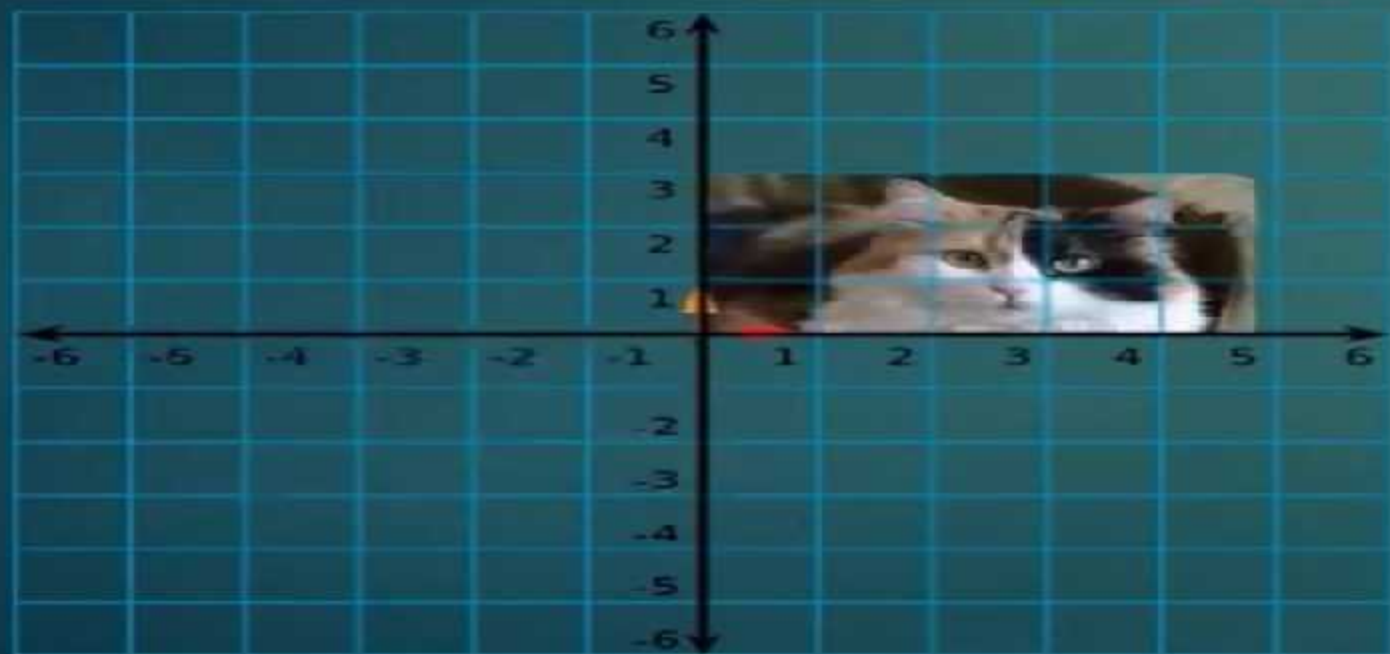
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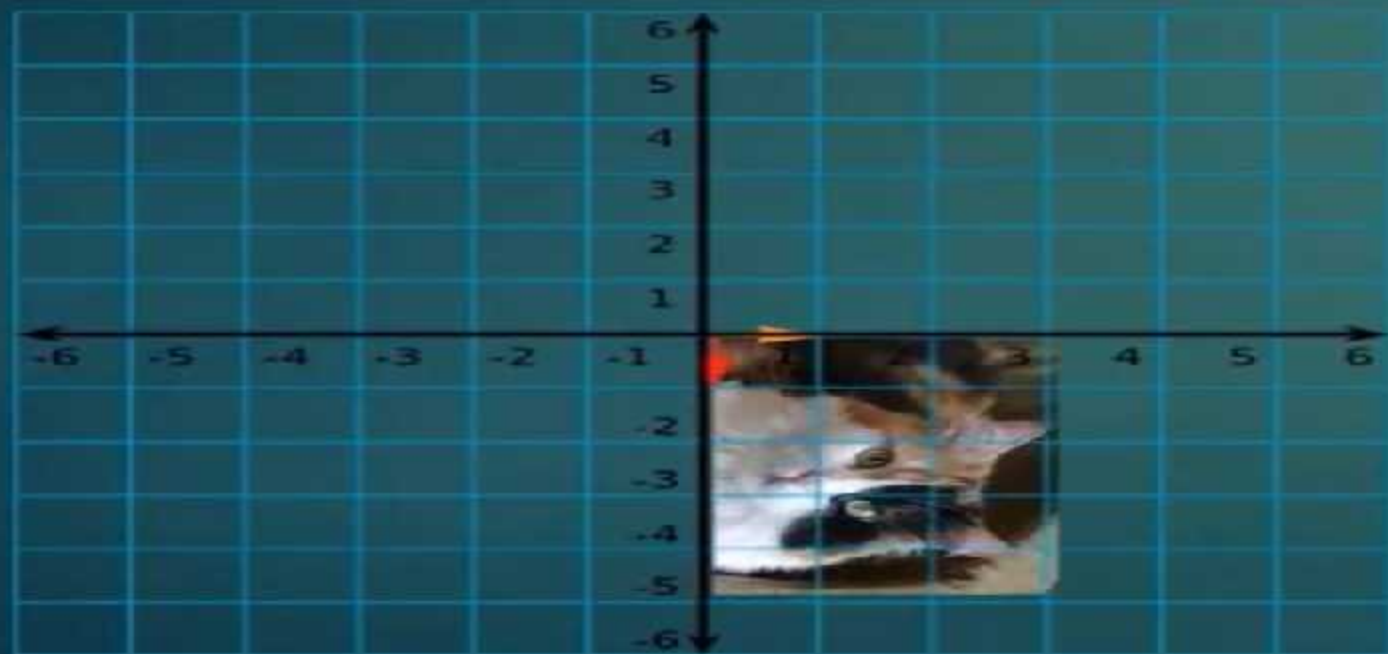
$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} x$$



$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{x}$$



$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{x}$$



$$\begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{x}$$



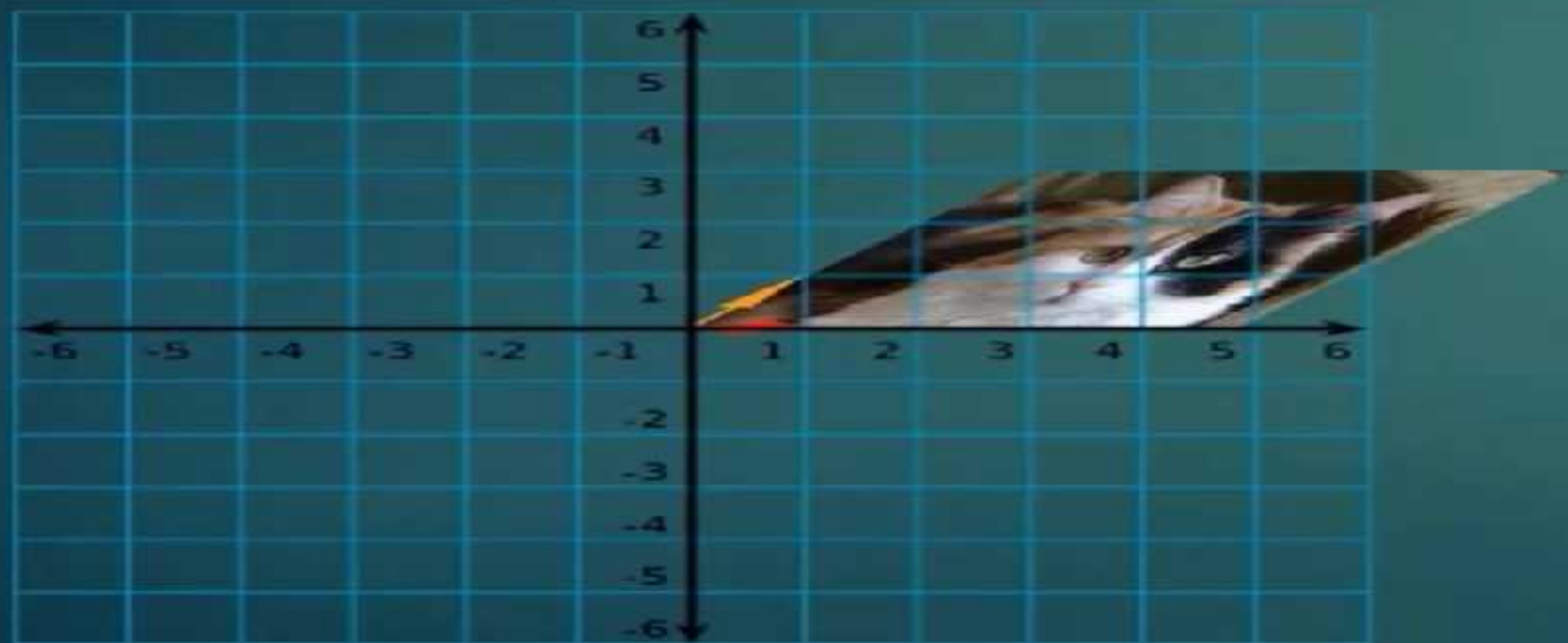
$$\begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{x}$$



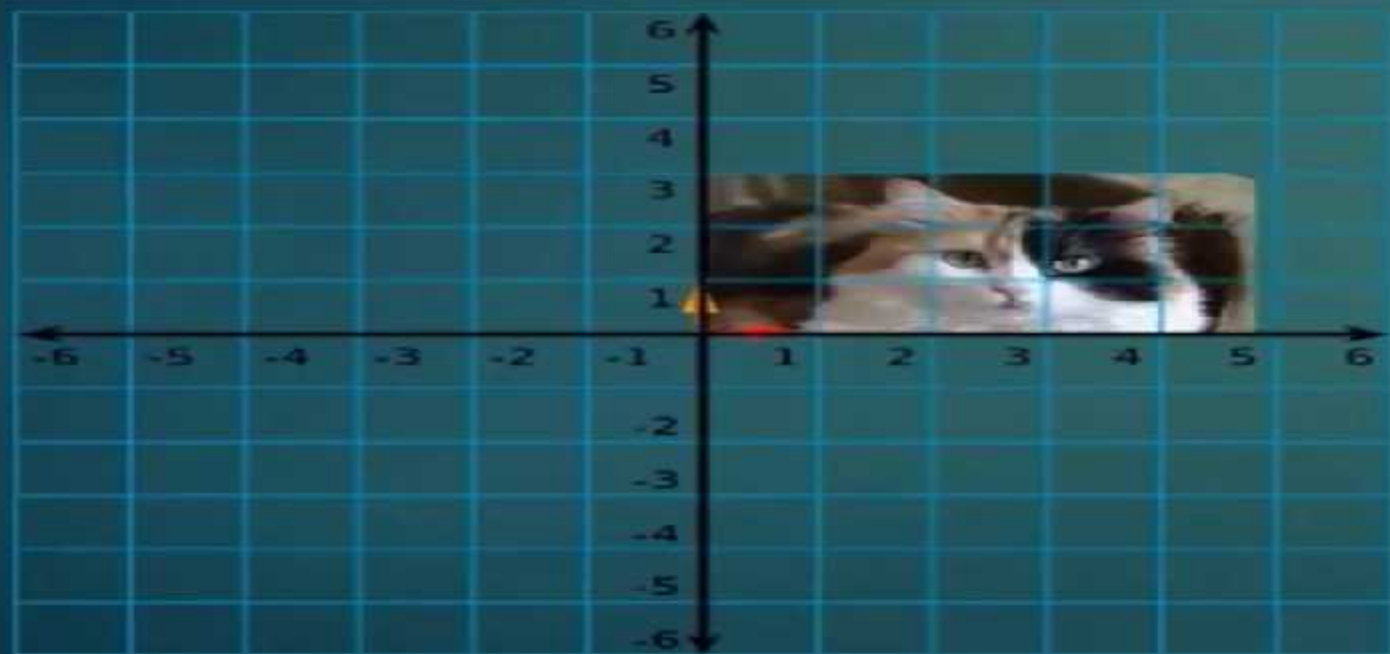
$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x}$$



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